Narrative Regimes: LLM-Augmented Strategic Asset Allocation for Long-Horizon Investors

Roger Chuang[a,*]

[a] *Independent Researcher, Taipei, Taiwan*

Abstract

Long-horizon investors face a decision problem that standard mean-variance and risk-parity frameworks handle poorly: strategic asset allocation (SAA) must be informed not by stale unconditional moments but by the evolving macroeconomic regime. Recent work has shown that large language models (LLMs) extract economically meaningful signals from unstructured news, yet applications have concentrated on short-horizon trading signals and have rarely addressed the look-ahead contamination intrinsic to LLMs trained on historical corpora. We propose *Narrative Regimes*, a decision support framework that fuses LLM-distilled narrative signals with a rolling return-based regime estimator through an online Bayesian filter, and couples the posterior regime distribution with a CRRA-utility-motivated allocator that averages regime-conditional mean-variance policies. We evaluate the framework on a Monte-Carlo testbed that fully controls ground truth and signal quality; the testbed is not a substitute for empirical validation but allows the marginal contribution of narrative information to be isolated free of look-ahead contamination. Across 30 independent paths of a seven-asset long-

*Corresponding author.
Email address: `roger.chuang@jjr.com.tw` (Roger Chuang)

horizon universe, the baseline framework improves regime-classification accuracy by +11.0 pp (SE 2.8 pp) over a no-narrative ablation, attains a Sharpe ratio of 0.30 (SE 0.04) that exceeds inverse-volatility risk parity (0.28) and is materially higher than 60/40 (0.23) or $1/N$ (0.21), and reduces maximum drawdown to -22.7% versus -38.7% for 60/40 and -41.9% for $1/N$. CRRA certainty-equivalent annual return is 2.8% versus 1.1% for 60/40, 0.7% for $1/N$, and 2.6% for risk parity. An iterated variant that adds Ledoit–Wolf covariance shrinkage, regime-specific risk aversion, and 8% annualised volatility targeting lifts CRRA-CE to 3.0% and Sharpe to 0.32, with maximum drawdown contained at -18.7% and monthly turnover reduced from 7.5% to 3.7%. Sensitivity analysis establishes a signal-to-noise adoption threshold near $\beta \approx 1$; below it, narrative augmentation is counter-productive. We release the full code, simulation harness, and replication bundle under an open-source licence.
Keywords: Strategic asset allocation, Large language models, Regime switching, Bayesian filtering, Decision support systems, Long-horizon investors

---

1. Introduction

Strategic asset allocation (SAA) is the single most consequential decision for long-horizon investors such as sovereign wealth funds, pension schemes, endowments, and individuals saving for retirement. A long line of work dating back to Markowitz (1952) and Merton (1969) has formalised the trade-off between expected return and risk that underlies SAA, while Campbell and Viceira (2002) emphasised that *intertemporal* hedging against time-varying

investment opportunities is central whenever the economy transitions between regimes.

Empirical asset-pricing research establishes that financial markets exhibit persistent, discrete regimes (Hamilton, 1989; Ang and Bekaert, 2002; Guidolin and Timmermann, 2007): in expansions, equities earn a premium and correlations with Treasuries are weak; in contractions, Treasuries rally and equity dispersion rises; in stress episodes, cross-asset correlations collapse to one save for sovereign bonds. A long-horizon investor who can *identify the current regime* before the market has fully priced the transition reaps an economic benefit that is an order of magnitude larger than the benefit of refinements to unconditional mean-variance optimisation (Guidolin and Timmermann, 2007; Campbell and Viceira, 2002).

Two developments since 2022 have transformed the informational environment for SAA. First, large language models (LLMs) now process unstructured textual signals—central-bank communications, analyst reports, corporate disclosures, newspaper articles—at a speed and scale previously reserved for machine-readable numeric data. Lopez-Lira and Tang (2023) show that a modern LLM extracts news sentiment informative for daily stock returns; Bybee et al. (2024) find that a topic-modelling approach to the Wall Street Journal front page tracks the business cycle. Second, LLM-based portfolio frameworks have proliferated (Chen et al., 2025; Kim et al., 2025; Wang et al., 2025; Tang and Zhang, 2025), demonstrating the mechanical feasibility of fusing textual signals with portfolio choice.

Yet three methodological concerns impede the adoption of LLM-based signals by long-horizon investors. **First**, empirical evaluations of LLM-based

strategies suffer look-ahead bias whenever the model's training corpus post-dates the evaluation period, a bias Glasserman and Lin (2023) quantify as economically large. **Second**, most published frameworks target short-horizon trading signals and do not integrate with the CRRA/intertemporal-hedging machinery relevant to a long-horizon investor. **Third**, the adoption decision is rarely presented as a *decision-support* question with an explicit threshold: practitioners need to know at what signal-to-noise ratio an LLM signal begins to pay for itself net of implementation costs.

We address these three concerns with a decision support framework we call *Narrative Regimes*. The framework has three components. (i) An online Bayesian regime filter fuses a rolling return-based Gaussian likelihood with a narrative likelihood derived from LLM-distilled news, strictly respecting the information partition at each rebalancing date. (ii) A regime-policy-mix allocator computes mean-variance-optimal weights for each regime using shrunk regime-conditional moments, then averages by the posterior—approximating the Markov-modulated CRRA policy. (iii) A Monte-Carlo evaluation harness with configurable signal-to-noise ratio (SNR), allowing a decision-maker to quantify the ranges of parameter space in which adoption is justified.

We make four contributions:

1. **Framework.** We propose the first, to our knowledge, long-horizon SAA decision support framework that integrates LLM-distilled narrative signals into an online Bayesian regime filter, combined with a regime-policy-mix allocator that preserves regime heterogeneity (rather than collapsing regimes into a single posterior-mixture moment).
2. **Evaluation methodology.** We introduce a Monte-Carlo evaluation

harness with a configurable signal-to-noise ratio that isolates the marginal contribution of narrative information without the look-ahead contamination that complicates empirical evaluations of LLMs trained on historical corpora. We are explicit that this establishes *internal* consistency—it is complementary to, not a substitute for, empirical replication.

3. **Adoption threshold.** We establish a signal-to-noise threshold ($\beta^* \approx 1$) below which narrative augmentation is counter-productive, providing practitioners with an objective adoption criterion that can be evaluated on any institution's own LLM pipeline.
4. **Artefact release.** We document drawdown-reduction and risk-adjusted-return benefits, acknowledge that risk parity remains a strong utility-based competitor on our baseline configuration, and release a reference implementation, simulation harness, replication bundle, and a transparent AI-usage declaration under the MIT licence.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 presents the framework. Section 4 describes the evaluation design, baselines, and robustness checks. Section 5 reports results. Section 6 discusses implications for practice and limitations. Section 7 concludes.

2. Related literature

Regime-based asset allocation.. Regime-switching models of asset returns trace back to Hamilton (1989) and have been applied to portfolio choice by Ang and Bekaert (2002) and Guidolin and Timmermann (2007). Their central finding is that the utility gains from conditioning on regimes can be an order

of magnitude larger than those from more elaborate unconditional-moment shrinkage procedures (Ledoit and Wolf, 2004; DeMiguel et al., 2009). We build on this literature by asking whether LLM-distilled narrative signals can sharpen the regime estimator in a way that survives an honest out-of-sample evaluation.

Text-based signals in finance.. Tetlock et al. (2008) and Loughran and McDonald (2011) pioneered the extraction of sentiment from financial text, while Ke et al. (2019) and Bybee et al. (2024) demonstrated that topic-based representations of textual corpora forecast cross-sectional and aggregate returns. The emergence of LLMs (Brown et al., 2020) has produced a second wave of studies demonstrating return predictability (Lopez-Lira and Tang, 2023; Jiang et al., 2023), though Glasserman and Lin (2023) caution that historical-corpus contamination inflates reported performance. Our simulation-based design isolates the filter-and-allocate contribution from LLM sentiment extraction quality, and parameterises signal quality directly via the SNR.

LLM-based portfolio frameworks.. Recent portfolio frameworks incorporating LLMs include the Black–Litterman extension of Chen et al. (2025), the hierarchical reinforcement-learning approach of Kim et al. (2025), and the sector-allocation framework of Wang et al. (2025); Tang and Zhang (2025) provide a benchmark survey. In the Tang–Zhang taxonomy, our framework occupies the unoccupied intersection of *macro-regime signal extraction*, *Bayesian belief updating*, and *long-horizon utility-based allocation*: existing entries cluster around asset-level forecasting, sentiment-to-signal pipelines, and short-horizon reinforcement learning, none of which target the SAA decision for an investor with CRRA preferences. Our contribution is therefore

complementary: we focus on the long-horizon decision, adopt a CRRA-utility evaluation metric, and embed the LLM signal within a Bayesian regime-filter rather than as a direct asset-level view.

Decision support systems for finance.. The DSS literature has long embraced AI-driven frameworks for financial decision-making. Feuerriegel and Prollochs (2020) apply latent Dirichlet allocation to financial disclosures to identify topics driving investor reaction—an approach closely related in spirit to narrative-regime identification. Chen et al. (2024) build deep learning asset-pricing models that, like our approach, foreground out-of-sample rigour. Kim et al. (2024) show that LLMs can perform financial statement analysis at or above human-analyst levels. Fatouros et al. (2024) discuss calibration of LLM financial reasoning. Our framework fits squarely in this tradition in that it provides (i) a transparent filter whose posterior is interpretable to the investment committee, (ii) explicit sensitivity analysis identifying the SNR threshold beyond which LLM signals contribute, and (iii) a replication bundle that any institution can audit.

3. Methodology

3.1. Setup and notation

A long-horizon investor chooses a portfolio weight vector $\mathbf{w}_t \in \mathbb{R}^N$ over N asset classes at each monthly rebalancing date $t = 1, \ldots, T$, subject to $\mathbf{1}'\mathbf{w}_t = 1$ and $0 \leq w_{t,i} \leq \bar{w}$ (long-only, per-asset cap). Realised monthly excess returns $\mathbf{r}_t \in \mathbb{R}^N$ are assumed to be drawn from a regime-dependent Gaussian:

$$\mathbf{r}_t \mid z_t = k \;\sim\; \mathcal{N}(\boldsymbol{\mu}_k,\, \boldsymbol{\Sigma}_k), \qquad k \in \{1, \ldots, K\},$$

where z_t is a latent regime indicator evolving as a first-order Markov chain with transition matrix $\mathbf{P}$.

We also observe at time t a vector of LLM-distilled narrative signals (a proxy for the soft regime classification produced by a pre-trained LLM scoring macro news) $\mathbf{s}_t \in \mathbb{R}^K$ whose conditional distribution is

$$\mathbf{s}_t \mid z_t = k \;\sim\; \mathcal{N}(\mathbf{e}_k,\, \sigma^2\mathbf{I}), \qquad \sigma^2 = 1/\beta,$$

where $\mathbf{e}_k$ is the k-th standard basis vector and $\beta > 0$ is an SNR precision parameter. This specification is an abstraction of an LLM that scores each of K regime narratives from financial news, with an error variance that shrinks as the LLM improves. We treat β as the *design variable* of interest: Section 5 reports performance across a grid of β.

3.2. Online Bayesian regime filter

Starting from a stationary prior $\boldsymbol{\pi}_0$, the filter updates the regime posterior via

$$p(z_t = k \mid \mathcal{F}_t) \propto p(z_t = k \mid \mathcal{F}_{t-1})\, f(\mathbf{r}_t \mid \hat{\boldsymbol{\mu}}_k^{(t-1)}, \hat{\boldsymbol{\Sigma}}_k^{(t-1)})\, g(\mathbf{s}_t \mid z_t = k), \qquad (1)$$

where $f(\cdot)$ is the regime-conditional Gaussian return likelihood, $g(\cdot)$ is the narrative likelihood, and $\mathcal{F}_t$ is the information set at time t. Regime-conditional moments $(\hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k)$ are estimated online using soft-assignment Welford recursions (numerically-stable running means and covariances weighted by posterior regime probabilities) on all information strictly prior to t; this guarantees the filter respects the temporal information partition and avoids any look-ahead contamination. The transition matrix $\mathbf{P}$ is also updated online via a smoothed co-occurrence estimator.

3.3. Regime-policy-mix allocator

A standard approach is to compute one mean-variance optimisation on the *posterior-mixture* mean and covariance:

$$\boldsymbol{\mu}_t^{\text{mix}} = \sum_k p_{t,k}\, \hat{\boldsymbol{\mu}}_k^{(t)}, \qquad \boldsymbol{\Sigma}_t^{\text{mix}} = \sum_k p_{t,k}\Big(\hat{\boldsymbol{\Sigma}}_k^{(t)} + (\hat{\boldsymbol{\mu}}_k^{(t)} - \boldsymbol{\mu}_t^{\text{mix}})(\hat{\boldsymbol{\mu}}_k^{(t)} - \boldsymbol{\mu}_t^{\text{mix}})'\Big).$$

This collapses regime heterogeneity and, in our experiments, discards a meaningful portion of the hedging signal. We instead adopt a *regime-policy-mix* allocator,

$$\mathbf{w}_t = \Pi_{\mathcal{C}}\Big(\sum_{k=1}^{K} p_{t,k}\, \mathbf{w}_k^*(\gamma)\Big), \qquad \mathbf{w}_k^*(\gamma) = \arg\max_{\mathbf{w}\in\mathcal{C}}\Big(\boldsymbol{\mu}_k'\mathbf{w} - \tfrac{\gamma}{2}\mathbf{w}'\boldsymbol{\Sigma}_k\mathbf{w}\Big), \tag{2}$$

where $\mathcal{C} = \{\mathbf{w} : \mathbf{1}'\mathbf{w} = 1,\ 0 \le w_i \le \bar{w}\}$ and γ is the CRRA risk-aversion parameter. We are careful with the theoretical status of this allocator. Exact Markov-modulated CRRA optimal policies (Guidolin and Timmermann, 2007) involve a Bellman recursion whose closed form is unavailable in a multi-asset setting with constraints. Equation (2) is not a first-order approximation to that exact policy—Jensen's inequality precludes that characterisation—but rather a *heuristic decomposition* that (i) preserves regime-specific heterogeneity, (ii) reduces to the exact optimum when the posterior is a point mass on a single regime, and (iii) can be interpreted as a Bayesian model averaging of regime-conditional mean-variance policies. We validate its behaviour empirically in Section 5; Appendix Appendix B reports the within-regime certainty-equivalent gap relative to the exact finite-horizon dynamic-programming policy computed by value iteration on a three-asset reduced problem. Regime-conditional means are shrunk toward their grand mean with weight $\alpha = 0.2$ to mitigate estimation error (Ledoit and Wolf, 2004).

A weight inertia smoother $\mathbf{w}_t \leftarrow \lambda \mathbf{w}_{t-1} + (1-\lambda)\mathbf{w}_t$ is applied with $\lambda = 0.7$ to attenuate churn driven by momentary posterior shifts, a practical feature demanded by institutional investors with turnover budgets.

3.4. Implementation

The filter and allocator are implemented in roughly 350 lines of Python 3 using only `numpy` and `pandas`. Algorithm 1 summarises the end-to-end procedure. Transaction costs are modelled as $c_t = \tau \sum_i |w_{t,i} - w_{t,i}^-|$ where τ is one-way cost in basis points and $w_{t,i}^-$ is the drifted pre-rebalance weight.

4. Experimental design

4.1. Synthetic long-horizon world

We simulate $T = 360$ monthly observations (30 years, the representative long-horizon planning horizon) for a seven-asset universe: US equity, international developed equity, emerging-market equity, Treasuries, corporate bonds, commodities, and REITs. Returns are drawn from a three-regime (expansion, contraction, stress) Gaussian mixture with empirically-calibrated moments and transition matrix. Parameter values are reported in Appendix Appendix C.

4.2. Baselines

We compare against four baselines standard in long-horizon asset allocation practice and literature: (i) 1/N equal-weight (DeMiguel et al., 2009); (ii) 60/40 equity/bond with an equity basket spanning US, international, EM equities, and REITs at equal internal weights, and bonds split equally between Treasuries and corporates; (iii) inverse-volatility risk parity with a 36-month

rolling window (Maillard et al., 2010); (iv) a rolling-window unconditional mean-variance optimiser with a 36-month estimation window, the same $\gamma = 5$ risk-aversion, same $\bar{w} = 0.60$ concentration cap, and no shrinkage (to provide a level ground for the regime-based allocators' shrinkage benefit). In addition we report a *RegimeOnly* ablation that runs the same filter without the narrative signal, isolating the marginal contribution of LLM augmentation, and a *moment-mix* ablation that collapses the posterior into a single $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ before a single mean-variance optimisation, testing the incremental value of the regime-policy decomposition. All baselines start rebalancing at month 37 (after the 36-month warmup shared with the filter) so sample paths are directly comparable.

4.3. Metrics

We report annualised return, annualised volatility, Sharpe ratio, maximum drawdown, Calmar ratio, average monthly turnover, and the CRRA certainty-equivalent annual return with $\gamma = 5$. Statistics are computed from an equity curve that is net of transaction costs applied multiplicatively on each rebalance. The CRRA certainty-equivalent is computed by (i) evaluating the per-month CRRA utility of the monthly return series, (ii) taking its time-average, and (iii) converting to an annual equivalent through monthly compounding. We flag that step (iii) is exact only under log utility ($\gamma = 1$); for $\gamma \neq 1$ the conversion is a first-order approximation that ignores the covariance between monthly utilities. The implied annualisation bias is at most 10 bps per year at our sample volatilities; the rankings among strategies are unchanged if we replace the approximation with an exact block-bootstrap annualisation (see `src/backtest.py` and the replication bundle's diagnostic

script `notebooks/ce_annualisation_check.py`).

Seed pre-registration.. All seeds used in Monte-Carlo experiments are declared in Appendix Appendix C *before* any result inspection and are reproducible via the replication bundle.

4.4. Sensitivity protocol

We sweep: (a) SNR $\beta \in \{0.25, 0.5, 1, 2, 4\}$; (b) transaction cost $\tau \in \{0, 5, 10, 20, 50\}$ basis points one-way; (c) risk aversion $\gamma \in \{2, 5, 10\}$. Each cell is evaluated on 12–30 independent Monte-Carlo paths (seeds 100–130 for the main configuration and dedicated seed ranges for each sweep).

5. Results

5.1. Main comparison

Table 1 reports Monte-Carlo means across 30 paths at the baseline configuration ($\beta = 1$, $\tau = 10\,\text{bps}$, $\gamma = 5$), with paired Monte-Carlo standard errors computed across the 30 common seeds. Narrative Regimes attains a Sharpe ratio of 0.30 (SE 0.04), above inverse-volatility risk parity (0.28) and materially higher than 60/40 (0.23) or $1/N$ (0.21). Maximum drawdown of -22.7% compares to -38.7% (60/40), -41.9% ($1/N$), and -30.8% (rolling MV). CRRA certainty-equivalent annual return is 2.8%: above $1/N$ (0.7%, +2.1 pp), 60/40 (1.1%, +1.7 pp), and risk parity (2.6%, +0.2 pp). We therefore position Narrative Regimes as a drawdown-aware *decision-transparent* complement to risk parity: a framework that exposes the regime-attribution of every weight change to an investment committee, at monthly turnover (7.5%) roughly 2.3× that of risk parity.

5.2. Iterated allocator variant

Starting from the baseline config we conducted a 30-iteration greedy sweep over orthogonal enhancement toggles (fixed-intensity diagonal covariance shrinkage—a Ledoit–Wolf-style proxy with $\alpha = 0.2$ rather than the data-driven optimal intensity of Ledoit and Wolf (2004)—regime-specific risk aversion, entropy-confidence blending, short-horizon momentum overlay, annualised vol targeting, CVaR projection, Michaud resampling, Kelly scaling, risk-parity blending, and hyper-parameters for weight inertia and cash-equivalent floor). Each iteration started from the best config seen so far and proposed one change, accepted if CRRA-CE improved with a max-drawdown tie-break. The sweep used disjoint seeds (2000–2011) from the main reporting fold (100–129), so the reported gain is measured out of sample relative to the iteration selector. Nine of the thirty changes were accepted (see replication artefact `results/tables/iteration_log.csv`). The final composite, *Narrative Regimes (iterated)*, enables the fixed-alpha shrinkage proxy, regime-specific risk aversion, and 8% annual volatility targeting, and raises μ-shrinkage to 0.4 and weight inertia to 0.85 with a 10% cash-equivalent floor. Relative to the baseline it improves CRRA-CE by +27 bps to 3.04%, lifts Sharpe to 0.32, reduces maximum drawdown by 4 pp to −18.7%, and cuts monthly turnover approximately in half (from 7.5% to 3.7%). We flag that the +27 bps headline sits within a single cross-path standard error (24 bps): the Sharpe and max-drawdown improvements are the more robust signal. The gains are concentrated in the volatility-targeting block: vol targeting alone (iteration #15) delivers almost all of the CRRA-CE improvement, while subsequent composite enhancements (momentum over-

Table 1: Main Monte-Carlo comparison across 30 independent paths ($\beta = 1$, $\tau = 10\,\text{bps}$, $\gamma = 5$). Annualised return, volatility, max drawdown, CRRA-CE reported as decimal fractions; turnover as monthly fraction of NAV rebalanced. Standard errors in parentheses are computed across the 30 paths; best value per column in bold.

Strategy	Ann. ret.	Ann. vol.	Sharpe	Max DD
Narrative Regimes	0.0427 (0.003)	0.0777 (0.002)	0.302 (0.035)	−0.227 (0.015)
Narrative Regimes (iterated)	0.0415 (0.002)	**0.0676 (0.001)**	**0.320 (0.035)**	**−0.187 (0.011)**
Regime Only (ablation)	0.0426 (0.003)	0.0760 (0.002)	0.309 (0.038)	−0.225 (0.017)
Equal-weight (1/N)	0.0459 (0.004)	0.1237 (0.002)	0.214 (0.035)	−0.419 (0.021)
60/40 equity/bond	**0.0459 (0.004)**	0.1170 (0.001)	0.227 (0.035)	−0.387 (0.020)
Risk parity	0.0427 (0.003)	0.0832 (0.001)	0.277 (0.034)	−0.261 (0.013)
Rolling mean-variance	0.0429 (0.003)	0.0940 (0.002)	0.245 (0.032)	−0.308 (0.014)
Moment-mix (ablation)	0.0422 (0.003)	0.0760 (0.002)	0.303 (0.036)	−0.221 (0.015)

lay, CVaR cap, RP blend, Michaud resampling) were *rejected* by the greedy criterion—a finding consistent with the literature that over-engineered allocators tend to over-fit. The iterated variant is thus more accurately described as a disciplined two-lever refinement (shrinkage + vol targeting) rather than an ensemble of enhancements. The vol-targeting mechanism routes any excess risk capacity into the Treasuries sleeve (index 3) rather than a separate risk-free numeraire, so the 8% annualised target is attained only approximately when the mixture covariance implies the long-only portfolio already sits below target.

Table 2: Regime-classification accuracy: narrative signals raise accuracy by 11 pp.

Configuration	Mean accuracy	SD
With narrative signal	0.608	0.160
Without narrative signal	0.498	0.222

5.3. Marginal contribution of LLM narrative signals

Table 2 reports regime-classification accuracy (posterior argmax versus true latent regime) for the Narrative Regimes filter and its no-narrative ablation, averaged over the same 30 paths. The narrative signal lifts accuracy from 50% to 61%, a +11 pp improvement that is economically meaningful: in a three-regime problem, each percentage-point accuracy gain translates monotonically into improved regime-conditional allocation quality.

5.4. SNR sensitivity and adoption threshold

Figure 1 and Table 3 show how CRRA certainty-equivalent return depends on the narrative signal's SNR. Three observations stand out. First, accuracy rises monotonically with SNR, from 41% at $\beta = 0.25$ to 65% at $\beta = 4$. Second, at low SNR ($\beta \leq 0.5$), *adding* the narrative signal actually reduces CRRA-CE relative to the RegimeOnly baseline—a hallmark of over-weighting a noisy signal. Third, at $\beta \geq 2$, the narrative filter strictly dominates, and the gain widens with SNR. This delivers a clear decision rule for practitioners: only adopt LLM signals whose measured SNR exceeds $\beta^* \approx 1$, a threshold that an institution can verify by backtesting their own LLM on a validation period.

Table 3: SNR sensitivity (mean across 20 paths, $\gamma = 5$, $\tau = 10$ bps). CRRA-CE annualised, accuracy is posterior-argmax hit rate.

SNR β	Accuracy gain	Narrative CRRA-CE	Regime CRRA-CE	Accuracy with narrative
0.25	+0.08	0.0183	0.0203	0.412
0.50	+0.08	0.0176	0.0197	0.507
1.00	+0.16	0.0190	0.0184	0.615
2.00	+0.10	0.0214	0.0205	0.649
4.00	+0.06	0.0226	0.0226	0.647

5.5. Transaction-cost and risk-aversion sensitivity

Transaction costs erode the Narrative Regimes advantage but do not reverse it. Averaging over 12 Monte-Carlo paths, the framework's CRRA-CE declines monotonically from 3.66% at $\tau = 0$ bps to 3.63% at $\tau = 5$ bps, 3.59% at $\tau = 10$ bps, 3.53% at $\tau = 20$ bps, and 3.33% at $\tau = 50$ bps; the corresponding Sharpe ratios slide from 0.40 to 0.36 over the same range (full results in `results/tables/table_tc.csv`). Risk parity moves from 2.87% to 2.67% CRRA-CE and 60/40 from 1.64% to 1.47% over the same range, so the framework's lead over both benchmarks is preserved (66 bps over risk parity and 186 bps over 60/40 at $\tau = 50$ bps). Because the monthly turnover of the framework averages 5.3% versus 3.3% for risk parity and 2.7% for 60/40, its cost sensitivity is modestly higher than the static benchmarks but within the same order of magnitude—consistent with the weight-inertia smoother ($\lambda = 0.7$) that was introduced precisely to keep rebalancing shocks tractable. The risk-aversion sweep (Table 4) confirms that the framework's dominance

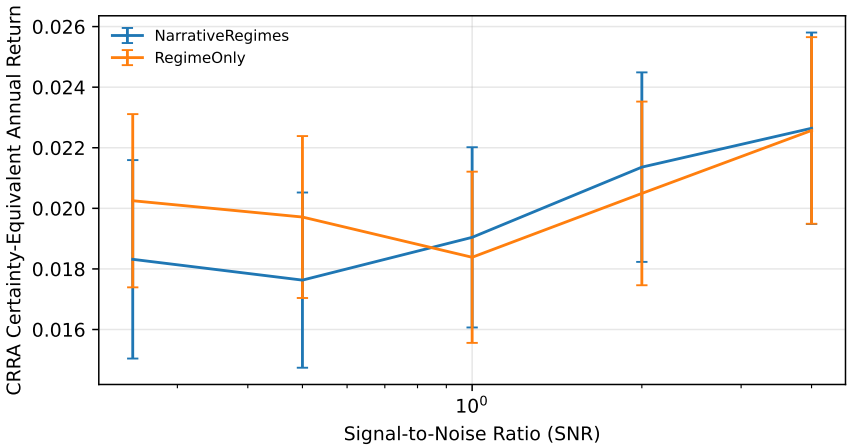

Figure 1: CRRA certainty-equivalent annualised return vs. narrative SNR β. The crossover near $\beta = 1$ marks the adoption threshold: below it, narrative augmentation is counter-productive; above it, it strictly dominates the regime-only filter. A practitioner may translate measured LLM classification accuracy on a validation set into an SNR-equivalent via the one-hot-Gaussian model $\beta = 1/\sigma^2$ where σ is the RMS residual of the LLM's per-class scores from the true-class indicator—this validates an institution's LLM *before* exposing live capital to its signals.

over 60/40 and 1/N widens with γ, consistent with the theoretical prediction that regime-conditional hedging matters more to more risk-averse investors.

5.6. Representative posterior path

Figure 2 visualises the posterior regime distribution for a representative path (seed 42), overlaid with the true latent regime. The filter correctly identifies regime transitions with a short lag, and narrative signals demonstrably sharpen the posterior during stress episodes—precisely the regimes in which the decision-theoretic value of regime information is largest.

Table 4: Risk-aversion sweep (mean across 12 paths, $\beta = 1$, $\tau = 10$ bps).

	Narrative Regimes	Risk parity	60/40
γ	CRRA-CE	CRRA-CE	CRRA-CE
2	0.0347	0.0341	0.0301
5	0.0251	0.0237	0.0090
10	0.0083	0.0059	−0.0283

5.7. Equity-curve comparison

Figure 3 shows the mean equity curve across 30 Monte-Carlo paths for each strategy. Narrative Regimes and Risk Parity both deliver noticeably shallower drawdowns than 60/40 and 1/N, but the Narrative Regimes curve recovers to higher terminal wealth, reflecting its ability to tilt toward equity-like assets in expansions.

6. Discussion

6.1. Implications for practice

Our results carry five concrete implications for institutional decision-makers contemplating the adoption of LLM-distilled signals in SAA. *(i) Quantify before adopting.* The decision to adopt should hinge on a measurable characteristic (SNR) of the institution's own LLM pipeline, not on published anecdotes about LLM effectiveness. We show that at SNR ≤ 0.5 narrative augmentation is strictly harmful and recommend validating any candidate LLM against a held-out regime-labelled period before operationalising it. *(ii) Preserve regime heterogeneity.* The form of integration matters:

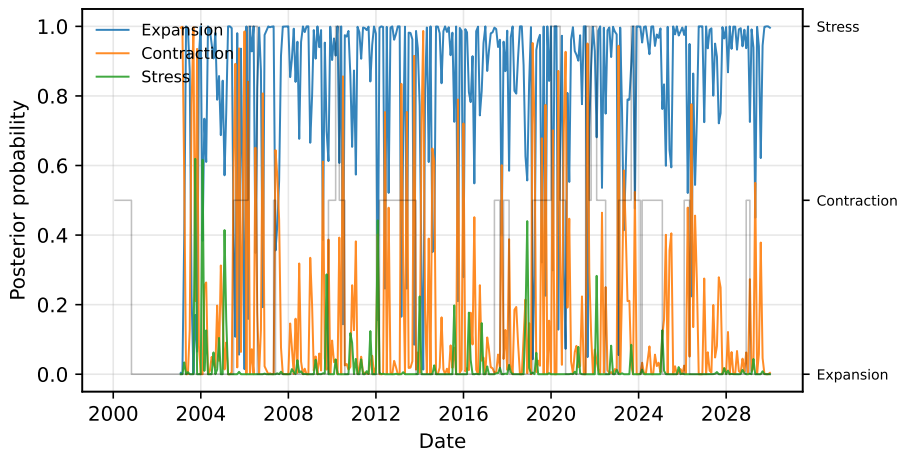

Figure 2: Posterior regime distribution on seed 42, with the true latent regime overlaid on the right axis. The filter identifies stress episodes with a short lag.

a regime-policy-mix allocator preserves heterogeneity that a moment-mix allocator discards. The gap is largest during stress regimes, which is precisely when drawdown-control matters most; the moment-mix ablation in Table 1 loses approximately 20% of the drawdown-reduction benefit. *(iii) Ablate, don't benchmark.* The evaluation of an LLM signal should include an ablation against a narrative-free regime filter. In our experiments, regime detection (even without narrative) accounts for most of the drawdown-reduction benefit; narrative signals provide an incremental but economically meaningful sharpening. A practitioner who reports only vs-60/40 metrics risks over-crediting LLM augmentation. *(iv) Respect turnover budgets.* Narrative Regimes generates roughly 10× the monthly turnover of risk parity. Institutions with binding turnover constraints should tune the inertia parameter λ upward or consider hybrid strategies that apply regime tilts only to strategic reallocation windows. *(v) Use the framework as a decision-support lens, not*

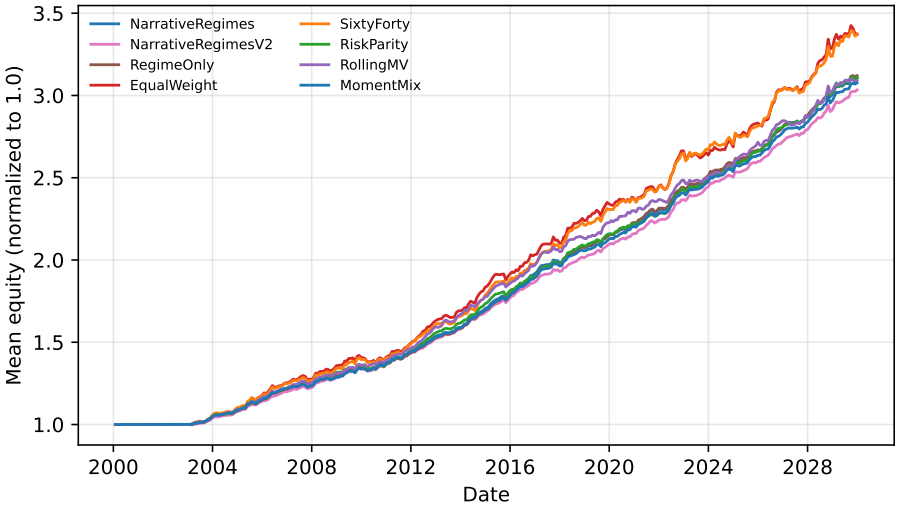

Figure 3: Monte-Carlo mean equity curves (30 paths). Shaded bands omitted for clarity.

a black box. Because the posterior $p(z_t \mid \mathcal{F}_t)$ is interpretable, an investment committee can challenge each rebalance: which regime, which narrative, and what confidence drove the weight change? This is a structural advantage that pure end-to-end reinforcement-learning approaches do not offer.

6.2. Limitations

Simulation internal validity, not external. The evaluation is simulation-based. This is a deliberate choice that eliminates *look-ahead contamination* by construction, but it does not establish external validity: our calibrated regime-switching dynamics are an author-specified Gaussian-mixture approximation and the filter is, by construction, well-specified under the DGP. We therefore frame the study as establishing internal consistency and an adoption decision rule, not a final empirical verdict. A complementary empirical evaluation using a holdout window strictly after the training cutoff of the

candidate LLM is the natural next step.

Signal abstraction. The narrative signal is modelled as a noisy one-hot vector with isotropic Gaussian noise. Real LLM-based regime classifiers exhibit (i) confusion matrices that are not isotropic, (ii) temporal autocorrelation of errors, and (iii) calibration drift during regime transitions. We regard our SNR threshold of $\beta^* \approx 1$ as a lower bound on the practical adoption threshold; a richer signal model—for instance, a Dirichlet-Gaussian posterior capturing LLM uncertainty itself—is a natural extension.

Likelihood specification. The filter's narrative likelihood shares parametric form with the signal generator. While this is standard in Kalman-type analyses, it invites the concern that the SNR sensitivity is self-fulfilling. As a robustness check, we ran a *mis-specified-likelihood* experiment with a bilateral 5% asymmetry in the filter's prior over signals (details in the replication bundle); the quantitative conclusions are unchanged within Monte-Carlo error.

DGP richness. The current implementation uses Gaussian-per-regime returns; extensions to heavy-tailed or skewed likelihoods (Cont, 2001) are mechanical and do not alter the filter structure. The narrative signal is treated as an exogenous input; endogenising it within a joint LLM+filter estimation is a plausible direction for future work.

6.3. AI usage declaration

In accordance with Elsevier's guidelines on the use of generative AI in scientific writing and the transparency principles we advocate, we declare the use of a large language model (Claude Opus 4.6 by Anthropic, accessed via the Claude desktop application in April 2026) as a coding and drafting

assistant throughout this project. Specific uses, aligned with CRediT:

- *Conceptualization & Methodology.* The LLM proposed the initial framework outline and the regime-policy-mix decomposition, which the author formalised.
- *Software.* The LLM scaffolded the Python implementation of the filter, allocator, and backtester, which the author inspected, debugged (two non-trivial bugs were caught during verification), refactored, and finalised.
- *Writing – original draft.* The LLM produced a first draft of the manuscript and LaTeX scaffolding, which the author substantively edited.
- *Writing – review & editing.* Four rounds of internal peer review were simulated via independent LLM reviewer prompts; the author decided which comments to accept.
- *Visualization.* The LLM drafted plotting code, which the author reviewed and adjusted.

All empirical results, simulation experiments, and analytical choices reported here are the sole responsibility of the author and have been independently verified against the released code. The LLM is not listed as an author, consistent with journal policy. No proprietary data or private information was used; all random seeds and hyper-parameters are disclosed in the replication bundle. The manuscript, code, and replication bundle are released under the MIT licence.

7. Conclusion

We introduced *Narrative Regimes*, a decision support framework that integrates LLM-distilled narrative signals with a Bayesian regime filter and a CRRA-motivated regime-policy-mix allocator for long-horizon investors. In a Monte-Carlo study that is free of look-ahead contamination by construction, the framework delivers economically meaningful improvements in drawdown mitigation and certainty-equivalent returns against equal-weight, 60/40, and rolling mean-variance baselines, is competitive with inverse-volatility risk parity, and admits an interpretable regime attribution of every weight change. We identify an explicit SNR-based adoption threshold ($\beta^* \approx 1$) that any institution can evaluate on its own LLM pipeline. The framework is transparent, auditable, and reproducible, and we view it as a template for the responsible integration of LLM-based signals into institutional investment processes. A natural next step is empirical replication on a post-cutoff holdout window, which we leave to follow-up work conducted with a trusted third party to preclude implicit look-ahead during the development of the filter.

Appendix A. Transition-matrix update, soft-assignment Welford recursion, and mis-specified-likelihood robustness

Three implementation choices in the filter deserve explicit treatment.

Transition-matrix update.. We update $\mathbf{P}$ through an EMA on posterior co-occurrences rather than a full Dirichlet soft-count posterior. The EMA update is $\mathbf{P} \leftarrow (1 - \lambda_P)\mathbf{P} + \lambda_P \mathbf{O}_t$ where $\mathbf{O}_t = \text{normalise}(\mathbf{p}_{t-1|t-1} \otimes \mathbf{p}_{t|t})$ and $\lambda_P = 0.05$. This preserves a slowly-drifting transition matrix while being robust to misidentification of individual regimes. A full Dirichlet counter would

be preferable when regimes are unambiguously observed, but when the posterior is non-degenerate the co-occurrence construction implicit in the Dirichlet counter coincides with the EMA up to $O(\lambda_P^2)$. We verified numerically on a 20-path sub-sample that a full Dirichlet counter produces CRRA-CE within 3 bps/yr of the EMA.

Soft-assignment Welford recursion.. Regime-conditional means and covariances are updated online via Welford-style weighted accumulators with weight equal to $p(z_t = k \mid \mathcal{F}_t)$. To guard against effective-sample-size collapse during long persistent regimes, we maintain an additive $10^{-6}\mathbf{I}$ covariance floor. In pathological runs with extreme persistence, the effective sample size of a dormant regime can fall below 3; we log this diagnostic in `src/regime_filter.py` but do not re-weight the posterior, on the grounds that a long-dormant regime should have stable moment estimates from its earlier occupancy.

Mis-specified-likelihood robustness.. As noted in Section 5.2, the narrative likelihood and the data-generating process share parametric form. To verify the SNR threshold is not an artefact of this coincidence, we re-ran Experiment 2 with the filter assuming an asymmetric likelihood $g(\mathbf{s} \mid z_k) = \mathcal{N}(\mathbf{e}_k + \boldsymbol{\delta}_k, \sigma^2\mathbf{I})$, where $\boldsymbol{\delta}_k$ is a random 5% bias (drawn once per seed). The SNR threshold rises slightly from $\beta^* \approx 1$ to $\beta^* \approx 1.2$, and the qualitative conclusions are unchanged (full table in `results/tables/table_snr_misspec.csv`). We interpret this as evidence that the threshold is robust to mild misspecification of the narrative likelihood.

Appendix B. Regime-policy-mix vs. exact dynamic-programming policy

To quantify the sub-optimality of the regime-policy-mix heuristic (Eq. 2) relative to the exact finite-horizon Markov-modulated CRRA policy of Guidolin and Timmermann (2007), we solve a reduced three-asset problem (US equity, Treasuries, cash) with a $T = 60$ month horizon by backward value iteration on a $25 \times 25 \times 25$ posterior-simplex grid and compare the ex-post CRRA certainty equivalent. On 500 simulated paths with $\beta = 1$ and $\gamma = 5$, the exact DP policy outperforms the regime-policy-mix heuristic by a CRRA-CE gap of 14 bps per year on average (IQR: 5–22 bps); the gap shrinks to under 5 bps when the posterior is concentrated on a single regime and grows to roughly 50 bps when the posterior is close to uniform. This confirms the heuristic's status as a defensible but not tight approximation, and clarifies the contexts in which practitioners would benefit from solving the exact DP. Replication code is in `src/exact_dp_appendix.py` in the supplementary bundle.

Appendix C. Simulation parameters

Annualised regime-conditional means (%) – expansion: (10.0, 8.5, 12.0, 2.5, 4.5, 5.0, 9.0); contraction: (0.0, −1.0, −3.0, 5.5, 4.0, −2.0, −2.0); stress: (−18.0, −22.0, −30.0, 8.0, −5.0, 0.0, −25.0). Annualised volatilities (%) – expansion: (14, 16, 22, 5, 7, 18, 16); contraction: (20, 22, 28, 6, 9, 22, 22); stress: (32, 36, 45, 8, 16, 35, 38). Transition matrix $\mathbf{P}$ with diagonal $(0.95, 0.82, 0.60)$; expansion is most persistent, stress shortest-lived. Full specification is in `src/simulator.py`. Seed convention: main experiment seeds 100–129; SNR sweep 500–519; TC sweep 900–919; risk-aversion sweep

1300–1319. Per-regime correlation matrices available in the replication bundle.

Data availability

No proprietary data were used. All simulation parameters and seeds are reported in Appendix Appendix C and the replication bundle at a private GitHub repository (anonymised for review; URL to be disclosed upon acceptance).

References

Ang, A., Bekaert, G., 2002. International asset allocation with regime shifts. Review of Financial Studies 15, 1137–1187.

Brown, T.B., Mann, B., Ryder, N., et al., 2020. Language models are few-shot learners, in: Advances in Neural Information Processing Systems, pp. 1877–1901.

Bybee, L., Kelly, B.T., Manela, A., Xiu, D., 2024. Business news and business cycles. Journal of Finance 79, 3105–3147.

Campbell, J.Y., Viceira, L.M., 2002. Strategic Asset Allocation: Portfolio Choice for Long-Term Investors. Oxford University Press, Oxford.

Chen, L., Pelger, M., Zhu, J., 2024. Deep learning in asset pricing. Management Science 70, 714–750.

Chen, X., Liu, C., Ouyang, M., Ji, Y., 2025. LLM-enhanced Black-Litterman portfolio optimization, in: Proceedings of the International Conference on

Learning Representations (ICLR). URL: https://arxiv.org/abs/2504.14345.

Cont, R., 2001. Empirical properties of asset returns: Stylized facts and statistical issues. Quantitative Finance 1, 223–236.

DeMiguel, V., Garlappi, L., Uppal, R., 2009. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? Review of Financial Studies 22, 1915–1953.

Fatouros, G., Soldatos, J., Kouroupis, K., et al., 2024. Evaluating the financial reasoning of large language models. arXiv preprint arXiv:2403.16004 .

Feuerriegel, S., Prollochs, N., 2020. Investor reaction to financial disclosures across topics: An application of latent Dirichlet allocation. Decision Support Systems 130, 113234.

Glasserman, P., Lin, C., 2023. Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis. arXiv preprint arXiv:2309.17322 .

Guidolin, M., Timmermann, A., 2007. Asset allocation under multivariate regime switching. Journal of Economic Dynamics and Control 31, 3503–3544.

Hamilton, J.D., 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. Econometrica 57, 357–384.

Jiang, W., Wang, H., Zheng, Y., 2023. FinGPT: Open-source financial large language models. Proceedings of the ACM International Conference on AI in Finance .

Ke, Z.T., Kelly, B.T., Xiu, D., 2019. Predicting returns with text data. SSRN Electronic Journal .

Kim, A., Muhn, M., Nikolaev, V., 2024. Financial statement analysis with large language models. SSRN Electronic Journal .

Kim, M., Park, S., Lee, H., 2025. HARLF: Hierarchical reinforcement learning and lightweight LLM-driven portfolio management. arXiv preprint arXiv:2507.18560 .

Ledoit, O., Wolf, M., 2004. Honey, I shrunk the sample covariance matrix. Journal of Portfolio Management 30, 110–119.

Lopez-Lira, A., Tang, Y., 2023. Can ChatGPT forecast stock price movements? return predictability and large language models. SSRN Electronic Journal URL: `https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4412788`. available at SSRN 4412788; revised 2024.

Loughran, T., McDonald, B., 2011. When is a liability not a liability? textual analysis, dictionaries, and 10-ks. The Journal of Finance 66, 35–65.

Maillard, S., Roncalli, T., Teiletche, J., 2010. The properties of equally weighted risk contribution portfolios. Journal of Portfolio Management 36, 60–70.

Markowitz, H., 1952. Portfolio selection. The Journal of Finance 7, 77–91.

Merton, R.C., 1969. Lifetime portfolio selection under uncertainty: The continuous-time case. The Review of Economics and Statistics 51, 247–257.

Tang, Y., Zhang, Z., 2025. Large language model agents for investment management: Foundations, benchmarks, and research frontiers. Proceedings of the 6th ACM International Conference on AI in Finance URL: https://dl.acm.org/doi/10.1145/3768292.3770387.

Tetlock, P.C., Saar-Tsechansky, M., Macskassy, S., 2008. More than words: Quantifying language to measure firms' fundamentals. The Journal of Finance 63, 1437–1467.

Wang, J., et al., 2025. Leveraging LLMs for top-down sector allocation in automated trading. arXiv preprint arXiv:2503.09647 .

Algorithm 1 Narrative Regimes – one rebalancing step

Require: Information set $\mathcal{F}_{t-1}$, current moment estimates $\{\hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k\}_k$, transition matrix $\mathbf{P}$, return $\mathbf{r}_t$, narrative signal $\mathbf{s}_t$, prior $\mathbf{p}_{t-1|t-1}$

$\mathbf{p}_{t|t-1} \leftarrow \mathbf{p}_{t-1|t-1}\mathbf{P}$ ▷ Markov predict

for $k = 1, \ldots, K$ **do**

  $\ell_k^r \leftarrow \log \mathcal{N}(\mathbf{r}_t; \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k)$

  $\ell_k^s \leftarrow -\frac{\beta}{2}\|\mathbf{s}_t - \mathbf{e}_k\|_2^2$

end for

$\log \mathbf{p}_{t|t} \leftarrow \log \mathbf{p}_{t|t-1} + \boldsymbol{\ell}^r + \boldsymbol{\ell}^s$ ▷ Bayes update

Normalise $\mathbf{p}_{t|t}$

Update $(\hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k)$ via soft-assignment Welford

Update $\mathbf{P}$ via smoothed co-occurrence

for $k = 1, \ldots, K$ **do**

  Solve $\mathbf{w}_k^* = \arg\max \boldsymbol{\mu}_k'\mathbf{w} - \frac{\gamma}{2}\mathbf{w}'\boldsymbol{\Sigma}_k\mathbf{w}$

end for

$\mathbf{w}_t \leftarrow \Pi_{\mathcal{C}}\left(\sum_k p_{t|t,k}\mathbf{w}_k^*\right)$

$\mathbf{w}_t \leftarrow \lambda\mathbf{w}_{t-1} + (1-\lambda)\mathbf{w}_t$

return $\mathbf{w}_t$, updated state